Assignment 2

Detection and Diversity in AWGN and Fading Channels

Submission deadline: *11:30 pm, 31st October, 2026*

General Instructions

- This assignment has three problems. Problem 1 is based on the lecture notes *Performance in Narrowband Fading Channels* [1]; Problems 2 and 3 are based on the lecture notes *Diversity in Narrowband Fading Channels* [2]. The schemes are described in the notes and are not repeated here; the relevant sections are cited in each problem. Complete all three problems.
- Read the Getting Started section first. It explains how a Monte Carlo simulation of a communication link is built.
- Submit **one consolidated report** (**PDF**) covering all tasks, together with the **code for each problem** and a `README` (see the end of this document). Each problem lists its own deliverables as a reminder.
- You may use Python or MATLAB.
- **Do not** use built-in functions that perform modulation, demodulation, detection or error-rate calculation (e.g. MATLAB's `qammod`, `pskdemod`, `berawgn`, `berfading`). You must implement these yourself. Standard numerical functions, such as random-number generators and `erfc` (`scipy.special.erfc` in Python), may be used.

Notation

The notation follows the lecture notes. The transmitted symbol is $x \in \mathcal{X} = \{u_1, \ldots, u_M\}$; the noise is w; the channel coefficient is h, and $\mathbf{h} = (h_1, \ldots, h_L)^\top$ when there are L antennas. SNR denotes the average signal-to-noise ratio, and $P_\mathrm{e} = \Pr[\hat{x} \neq x]$ the symbol error probability. In this assignment, 4-QAM refers to the constellation with points at $\pm A \pm jA$, also known as QPSK.

Getting Started

The performance of a digital communication system can be evaluated by simulating its baseband processing chain end to end, as shown in Fig. 1. In this assignment, you will estimate the symbol error rate (SER) of a given modulation scheme and detector empirically, by Monte Carlo simulation.

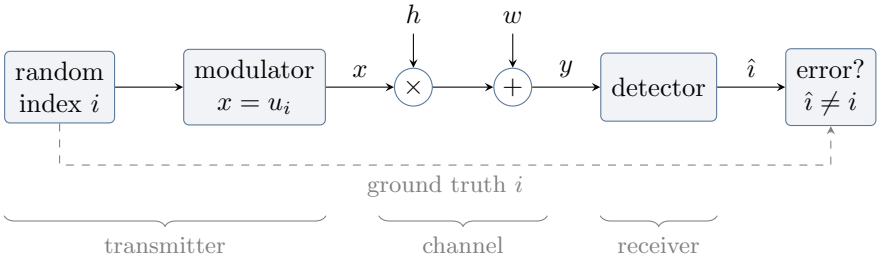


Figure 1: The simulated baseband communication link. The dashed line from h to the detector indicates that the channel is known at the receiver.

The transmitter selects a symbol x from a constellation and sends it through the channel. The receiver then decides on $\hat{x}$ based on the received signal. The quantity of interest is the

symbol error probability

$$P_{\mathrm{e}} = \Pr[\hat{x} \neq x],$$

which depends on the constellation, the channel, the SNR and the detector. Repeating this experiment over many independent realizations of the symbol, the channel and the noise, and recording the fraction of incorrect decisions, yields an estimate of P_{e}. This methodology is called Monte Carlo simulation.

G.1 Transmitter

The transmitted symbol x belongs to a constellation $\mathcal{X} = \{u_1, \ldots, u_M\}$, where M is the modulation order and each u_i is a point in the complex (I–Q) plane. Since the symbols are equiprobable, the transmitter draws an index i uniformly from $\{1, \ldots, M\}$ and sends $x = u_i$. The index i serves as the ground truth against which the receiver's decision is compared.

Construct the constellation so that its average symbol energy is unity:

$$\mathbb{E}\big[|x|^2\big] = \frac{1}{M}\sum_{i=1}^{M} |u_i|^2 = 1.$$

With this normalization, the SNR is controlled entirely by the noise variance, as described in G.2. The constellation therefore does not need to be regenerated for each SNR, and different modulation schemes are compared on an equal-energy basis.

G.2 SNR and the Channel Model

For both the additive white Gaussian noise (AWGN) channel ($h = 1$) and the flat-fading channel, the received baseband signal is modelled as

$$y = hx + w.$$

The average SNR is the ratio of the average symbol energy to the noise variance,

$$\mathrm{SNR} = \frac{\mathbb{E}\big[|x|^2\big]}{\mathbb{E}\big[|w|^2\big]}.$$

For the fading channel, $\mathbb{E}\big[|h|^2\big] = 1$, so this is also the average received SNR.

Given a target SNR in decibels, $\mathrm{SNR}_{\mathrm{dB}}$, the linear ratio is $\mathrm{SNR} = 10^{\mathrm{SNR}_{\mathrm{dB}}/10}$. Since $\mathbb{E}\big[|x|^2\big] = 1$, the SNR is set by the noise variance alone: generate the noise as

$$w \sim \mathcal{CN}\big(0, \tfrac{1}{\mathrm{SNR}}\big)\,.$$

Use this convention consistently in all your simulation scripts.

G.3 Generating Channel and Noise Realizations

For every transmitted symbol, draw a fresh, independent realization of the noise w and of the channel coefficient h from their respective distributions.

- **Additive noise:** $w \sim \mathcal{CN}(0, 1/\mathrm{SNR})$, i.e. $w = \frac{1}{\sqrt{2\,\mathrm{SNR}}}\,(w_I + j w_Q)$, where $w_I, w_Q \sim \mathcal{N}(0,1)$ are independent.
- **Rayleigh fading coefficient:** $h \sim \mathcal{CN}(0,1)$, i.e. $h = \frac{1}{\sqrt{2}}(h_I + j h_Q)$, where $h_I, h_Q \sim \mathcal{N}(0,1)$ are independent of each other and of the noise.

Generate the real Gaussian samples with the standard normal generator of your language (`randn` in MATLAB, `numpy.random.Generator.standard_normal` in Python).

G.4 Receiver: Detection

The detector computes an estimate $\hat{x} = u_{\hat{\imath}}$ from the received signal y. Use the maximum-likelihood (ML) detector, which for equiprobable symbols in Gaussian noise reduces to the minimum-distance rule. In the AWGN channel the rule is given in [1, Sec. 4.1]. In the fading channel the receiver knows h and performs coherent detection, as described in [1, Sec. 5.1].

Implement the detector for an *arbitrary* constellation, so that the same function serves BPSK and 4-QAM and is reused in Problems 2 and 3. A vectorized implementation computes, for all N received samples at once, the $N \times M$ array of distances to the (scaled) constellation points and takes the index of the minimum along the constellation dimension. The detector should return the index $\hat{\imath}$.

G.5 Counting Errors and Estimating P_{e}

Let $\mathbf{1}\{\hat{\imath} \neq i\}$ denote the error indicator, which equals 1 when the detector is wrong and 0 otherwise. Its mean is the error probability:

$$P_{\mathrm{e}} = \mathbb{E}\big[\mathbf{1}\{\hat{\imath} \neq i\}\big].$$

After sending N independent symbols and observing n_{err} errors, the sample mean

$$\hat{P}_{\mathrm{e}} = \frac{1}{N}\sum_{n=1}^{N} \mathbf{1}\{\hat{\imath}_n \neq i_n\} = \frac{n_{\mathrm{err}}}{N}$$

is the *symbol error rate* (SER). By the law of large numbers, it converges to P_{e} as $N \to \infty$. Count errors by comparing indices ($\hat{\imath} \neq i$) rather than complex values.

How many symbols? $\hat{P}_{\mathrm{e}}$ is itself a random variable. Each indicator is Bernoulli(P_{e}), so $\mathrm{std}(\hat{P}_{\mathrm{e}}) = \sqrt{P_{\mathrm{e}}(1-P_{\mathrm{e}})/N}$, and for $P_{\mathrm{e}} \ll 1$

$$\frac{\mathrm{std}(\hat{P}_{\mathrm{e}})}{P_{\mathrm{e}}} \approx \frac{1}{\sqrt{n_{\mathrm{err}}}}.$$

The accuracy of a Monte Carlo estimate is therefore set by the number of errors observed, not by the number of symbols sent. About 100 errors give an accuracy of approximately 10%, which is adequate for a logarithmic plot. In this assignment, $N = 10^6$ symbols are simulated per SNR point, so that error rates down to $P_{\mathrm{e}} \approx 10^{-4}$ are estimated reliably. The SNR ranges in each problem are chosen so that the curves reach this level.

G.6 The Monte Carlo Procedure

Algorithm 1 Monte Carlo estimation of P_{e}

Input: Unit-energy constellation $\mathcal{X} = \{u_1, \ldots, u_M\}$; array of $\mathrm{SNR}_{\mathrm{dB}}$ values; number of symbols $N = 10^6$

Fix the random seed

for each $\mathrm{SNR}_{\mathrm{dB}}$ in the array **do**

  $\mathrm{SNR} \leftarrow 10^{\mathrm{SNR}_{\mathrm{dB}}/10}$

  Draw N indices $[i_1, \ldots, i_N]$ uniformly from $\{1, \ldots, M\}$ and set $x_n = u_{i_n}$

  Generate the channel and the received samples: $([h_1, \ldots, h_N], [y_1, \ldots, y_N]) \leftarrow \mathrm{Channel}([x_1, \ldots, x_N], \mathrm{SNR})$

  Detect all samples: $[\hat{\imath}_1, \ldots, \hat{\imath}_N] \leftarrow \mathrm{Detector}([y_1, \ldots, y_N], [h_1, \ldots, h_N], \mathcal{X})$

  Count the errors: $n_{\mathrm{err}} \leftarrow \sum_{n=1}^{N} \mathbf{1}\{\hat{\imath}_n \neq i_n\}$

  $\hat{P}_{\mathrm{e}}(\mathrm{SNR}_{\mathrm{dB}}) \leftarrow n_{\mathrm{err}}/N$

end for

G.7 Plotting Performance Curves

Since the SER spans several orders of magnitude, plot it on a logarithmic vertical axis against $\mathrm{SNR}_{\mathrm{dB}}$ on a linear horizontal axis: use `plt.semilogy()` in Python or `semilogy` in MATLAB. Plot only the points with $n_{\mathrm{err}} \geq 10$; points with fewer errors are too inaccurate to be meaningful.

Problem 1. Symbol Error Rate in AWGN and Fading Channels

Consider the channel

$$y = hx + w, \qquad x \in \mathcal{X}, \qquad w \sim \mathcal{CN}(0, 1/\mathrm{SNR}). \tag{1}$$

Use the BPSK and 4-QAM constellations of [1, Sec. 2.2], each normalized to unit average energy, and the ML detector in all cases. Implement every block of Fig. 1 yourself, following Getting Started.

The simulated curves are to be compared with the following theoretical expressions, where $Q(a) = \frac{1}{2}\operatorname{erfc}(a/\sqrt{2})$ is the Gaussian tail function [1, Sec. 3.1]. For BPSK, they are derived in [1, Secs. 4.3 and 5.2]. For 4-QAM, they follow from the fact that the in-phase and quadrature components are detected independently, each as a BPSK symbol carrying half of the energy. In the fading channel, the 4-QAM expression is twice the BPSK result with SNR replaced by SNR/2; it neglects the small probability that both components are in error, and is therefore an approximation.

	AWGN ($h = 1$)	Rayleigh fading ($h \sim \mathcal{CN}(0,1)$)
BPSK	$P_{\mathrm{e}} = Q\big(\sqrt{2\,\mathrm{SNR}}\big)$	$P_{\mathrm{e}} = \frac{1}{2}\left(1 - \sqrt{\frac{\mathrm{SNR}}{1+\mathrm{SNR}}}\right)$
4-QAM	$P_{\mathrm{e}} = 2Q\big(\sqrt{\mathrm{SNR}}\big) - Q^2\big(\sqrt{\mathrm{SNR}}\big)$	$P_{\mathrm{e}} \approx 2 \cdot \frac{1}{2}\left(1 - \sqrt{\frac{\mathrm{SNR}/2}{1+\mathrm{SNR}/2}}\right)$

1.1 Task 1: AWGN channel

1. **Simulation.** Estimate the SER of BPSK and 4-QAM in the AWGN channel ($h = 1$) for $\mathrm{SNR}_{\mathrm{dB}} \in \{0, 1, \ldots, 12\}$.
2. **Theory.** Evaluate the AWGN expressions of the table above over the same SNR range.
3. **Plot.** Plot the simulated SER (markers) and the theoretical SER (lines) of both schemes versus $\mathrm{SNR}_{\mathrm{dB}}$ on a single figure, with a legend.

1.2 Task 2: Rayleigh fading channel

Consider the Rayleigh fading channel with $h \sim \mathcal{CN}(0, 1)$, drawn independently for every symbol and known at the receiver.

1. **Simulation.** Estimate the SER of BPSK and 4-QAM with coherent ML detection for $\mathrm{SNR}_{\mathrm{dB}} \in \{0, 2, \ldots, 40\}$.
2. **Theory.** Evaluate the Rayleigh-fading expressions of the table above over the same SNR range.
3. **Plot.** Plot the simulated and theoretical SER of both schemes versus $\mathrm{SNR}_{\mathrm{dB}}$ on a single figure, using the same conventions as in Task 1.

1.3 Deliverables

- **Code:** constellation generation, channel, ML detector, error counting, and the theoretical expressions (with $Q(\cdot)$ as a separate function).
- **Figures:** the AWGN figure of Task 1 and the fading figure of Task 2.
- **Discussion:**
  - the agreement between simulation and theory, and the reason for any deviation;
  - the qualitative difference between the AWGN and fading curves at high SNR, explained using the deep-fade argument of [1, Sec. 5.4].

Problem 2. Receive Diversity

Consider a transmitter with one antenna and a receiver with L antennas, the single-input multiple-output (SIMO) channel of [2, Sec. 3]. The received signal is

$$\mathbf{y} = \mathbf{h}\,x + \mathbf{w}, \qquad \mathbf{h} = (h_1, \ldots, h_L)^\top, \qquad \mathbf{w} = (w_1, \ldots, w_L)^\top, \tag{2}$$

where the channel coefficients $h_\ell \sim \mathcal{CN}(0,1)$ are i.i.d. across antennas, and the noise samples $w_\ell \sim \mathcal{CN}(0, 1/\mathrm{SNR})$ are i.i.d. and independent of the channel. Draw a new $\mathbf{h}$ for every symbol. The receiver knows $\mathbf{h}$. The SNR is defined as in G.2 and is therefore the average SNR *per receive antenna*; for $L = 1$ the model reduces to the fading channel of Problem 1. Use 4-QAM throughout.

2.1 Task 1: Combining schemes

Implement the following two receivers.

- **Maximal ratio combining (MRC)** [2, Secs. 3.1–3.2]: the L received signals are combined into a scalar decision statistic, and the ML detector of Problem 1 is applied to it.
- **Selection combining (SC)** [2, Sec. 3.5]: only the antenna with the largest instantaneous channel gain $|h_\ell|^2$ is used, followed by the coherent ML detector of Problem 1 on that antenna.

In your report, state the decision statistic of each scheme as an equivalent scalar channel, giving its effective gain and its noise variance. If your MRC implementation does not normalize the combined signal by $\|\mathbf{h}\|$, show that the detector you apply to it is still equivalent to the ML rule.

2.2 Task 2: Simulation

Estimate the SER of MRC and SC for $L \in \{1, 2, 4\}$ and $\mathrm{SNR}_{\mathrm{dB}} \in \{0, 2, \ldots, 24\}$. Plot all curves on a single figure, using the same colour for a given L, solid lines for MRC and dashed lines for SC.

2.3 Deliverables

- **Code:** the multi-antenna channel and the MRC and SC receivers, reusing the constellation and detector of Problem 1.
- **Figure:** SER versus $\mathrm{SNR}_{\mathrm{dB}}$ for MRC and SC with $L = 1, 2, 4$.
- **Discussion:**

- the decision statistics of Task 1;
- a consistency check: the curves for $L = 1$ should coincide with each other and with the 4-QAM fading curve of Problem 1;
- the slope of each curve at high SNR, compared with the diversity order predicted in [2];
- the SNR gap between MRC and SC for $L = 2$ and $L = 4$, and an explanation of its origin in terms of array gain [2, Sec. 3.4].

Problem 3. Transmit Diversity

Consider a transmitter with $L = 2$ antennas and a receiver with a single antenna, the multiple-input single-output (MISO) channel of [2, Sec. 4]. Let $h_1, h_2 \sim \mathcal{CN}(0, 1)$ be the i.i.d. channel coefficients from the two transmit antennas, and let the noise be generated as in G.3. The receiver knows h_1 and h_2. Use 4-QAM throughout.

For a fair comparison, the *total* energy transmitted per channel use, summed over both antennas, must equal that of the single-antenna system, i.e. unity. The SNR is defined as in G.2, with the noise variance 1/SNR referred to this total transmitted energy.

3.1 Task 1: Transmission schemes

Implement the following three schemes.

- **Single antenna (reference):** the fading channel of Problem 1.
- **Transmit beamforming** [2, Sec. 4.1]: the transmitter knows (h_1, h_2) and sends the same symbol from both antennas, weighted so that the two components add coherently at the receiver.
- **Alamouti scheme** [2, Sec. 4.2]: the transmitter has no channel knowledge. Two symbols are sent over two consecutive channel uses using the Alamouti code. The channel is constant over the two channel uses of a block and independent across blocks; simulate $N/2$ blocks, so that N symbols are sent in total. The receiver performs Alamouti combining followed by ML detection of each symbol, and the SER is counted over all N symbols.

In your report, state the beamforming weights, the Alamouti transmission matrix, and the combined decision statistics. State explicitly the scaling applied to the transmitted symbols in each scheme to satisfy the total-energy constraint, and verify it numerically by computing the average transmitted energy per channel use in your simulation.

3.2 Task 2: Simulation

Estimate the SER of the three schemes for $\mathrm{SNR}_{\mathrm{dB}} \in \{0, 2, \ldots, 24\}$. Plot them on a single figure, together with the MRC curve for $L = 2$ from Problem 2.

3.3 Deliverables

- **Code:** the beamforming and Alamouti transmitters and receivers, reusing the constellation and detector of Problem 1.
- **Figure:** SER versus $\mathrm{SNR}_{\mathrm{dB}}$ for the single-antenna, beamforming and Alamouti schemes, and for MRC with $L = 2$.
- **Discussion:**
  - the diversity order of each scheme, as observed from the slope of its curve at high SNR;
  - the SNR gaps between beamforming, Alamouti and MRC with $L = 2$, and an explanation of their origin.

Submission Requirements

Submit a single archive containing:

1. **Report (PDF):** all figures and discussions for Problems 1–3. Every figure must have labelled axes, a legend, and a caption stating the SNR range and the number of Monte Carlo symbols per SNR point.
2. **Code:** complete source code, organized as below.
3. `README`**:** language and version; required packages (or MATLAB version and toolboxes); the command that reproduces each problem's figures; the random seed(s) used.

Coding requirements

- Keep constellation generation, channel generation, detection, error counting and the theoretical expressions in separate functions. All simulations must *reuse* the same constellation and detector code rather than copying it.
- **Vectorize the simulation over all symbols**, as in Algorithm 1. Do *not* write a loop that transmits, detects and checks one symbol at a time. Python and MATLAB execute such loops slowly, and this assignment requires about 10^6 symbols for each SNR point, each scheme and each value of L, which amounts to several hundred million loop iterations in total. A vectorized simulation of one curve runs in seconds. A loop can be over different SNR values.
- **All randomness must be reproducible.** Set the seed of the random-number generator once, at the start of each script (e.g. `rng = np.random.default_rng(seed)` in Python, `rng(seed)` in MATLAB), and state the seed in the `README`. Running the submitted code with this seed must reproduce the figures in the report exactly.

Suggested layout

```
assignment2/
|-- common/        constellation.py|m, channel.py|m, detector.py|m, theory.py|m
|-- problem1/      awgn_ser.py|m, rayleigh_ser.py|m
|-- problem2/      receive_diversity.py|m
|-- problem3/      transmit_diversity.py|m
|-- README
```

Grading note. The emphasis is on implementing the communication algorithms. A plot or number without the code that produced it receives no credit. A simulated curve that disagrees with theory, or with the corresponding results in the lecture notes, must be discussed, not hidden.

References

[1] R. Gangula, *Performance in Narrowband Fading Channels*, Lecture Notes, EC5.407 Wireless Communications, IIIT Hyderabad.

[2] R. Gangula, *Diversity in Narrowband Fading Channels*, Lecture Notes, EC5.407 Wireless Communications, IIIT Hyderabad.

End of Assignment